

MigrantPay

Agile Process Model Document

Team 20 | MIT Academy of Engineering | 22 May 2026

1. What is the Agile Process Model?

Agile is an iterative software development methodology that delivers working software in short cycles called Sprints or Iterations. Rather than completing the entire system before testing, Agile builds, tests, and reviews small increments of functionality repeatedly. The Agile Manifesto prioritises: Individuals and interactions over processes and tools; Working software over comprehensive documentation; Customer collaboration over contract negotiation; Responding to change over following a fixed plan. For MigrantPay, we adopt the Agile model with a Scrum-inspired structure adapted to the 24-hour hackathon timeline.

2. Why Agile for MigrantPay?

We selected Agile over Waterfall and Spiral models based on the following justifications:

Criterion	Waterfall	Spiral	Agile (Selected)
Development Style	Sequential, rigid	Risk-driven, complex	Iterative, flexible
Requirement Changes	Very hard to accommodate	Moderate	Easily accommodated
Team Size Fit	Large teams	Large, complex teams	Small teams (3-6)
Delivery Speed	Slow (full cycle before delivery)	Moderate	Fast (incremental delivery)
Testing Approach	End of project	Each phase	Continuous / parallel
Hackathon Compatibility	Poor	Moderate	Excellent
Documentation Flexibility	Rigid upfront docs	Heavy process docs	Lightweight, living docs
Feedback Integration	Not supported mid-project	Limited	Core principle

Agile is the ideal fit because: our team is small (4 members); requirements may evolve during the hackathon; we need to demonstrate a working prototype within 24 hours; judges may suggest changes during Review 1 which we can adapt in the next sprint; and module-based development aligns perfectly with Agile's incremental delivery approach.

3. MigrantPay Agile Sprint Plan

We divided the 24-hour hackathon into 4 sprints, each with defined deliverables, owners, and acceptance criteria.

Sprint	Duration	Goal	Deliverables	Owner
Sprint 0 (Planning)	0–1 hour	Project Setup & Planning	GitHub repo setup, project scaffolding, task assignment, tech stack finalization	All members
Sprint 1 (Foundation)	1–6 hours	Authentication & Wallet	Login/Register pages, OTP simulation, JWT setup, Wallet dashboard, Database schema	Member A + B
Sprint 2 (Core Features)	6–14 hours	Send Money & History	Transfer form, balance validation, transaction processing, transaction history page, receipt generation	Member C + D
Sprint 3 (Enhancement)	14–20 hours	Language + Tips + Polish	i18n integration (EN/HI/MR), financial tips component, UI polish, mobile responsiveness, error handling	All members
Sprint 4 (Review Prep)	20–24 hours	Documentation & Demo	SRS finalization, UML diagrams, test cases, COCOMO estimation, presentation slides, live demo preparation	All members

4. Agile Roles in MigrantPay

Agile Role	Team Member	Responsibilities
Product Owner	Member A	Define user stories, prioritize features, ensure MVP aligns with problem statement
Scrum Master	Member B	Coordinate sprints, remove blockers, ensure daily standup and sprint reviews
Developer (Frontend)	Member C	React UI, Tailwind CSS, multilingual components, mobile responsiveness
Developer (Backend)	Member D	Node.js API, Firebase integration, transaction logic, authentication

5. Agile Ceremonies Adapted for Hackathon

Ceremony	Frequency	Duration	Purpose
Sprint Planning	Before each sprint	15 min	Assign tasks, define sprint goal, estimate effort

Ceremony	Frequency	Duration	Purpose
Daily Standup	Every 3 hours	5 min	Share progress, identify blockers, adjust plan
Sprint Review	After each sprint	10 min	Demo completed features, get feedback
Sprint Retrospective	End of Sprint 2 and 4	5 min	Identify what worked, what to improve
Backlog Refinement	Sprint 0 and ongoing	10 min	Update task list based on changing priorities

6. MigrantPay Product Backlog

The product backlog lists all user stories prioritized by business value and sprint assignment.

ID	User Story	Priority	Story Points	Sprint
US-01	As a migrant worker, I want to register with my mobile number so I can access the platform.	High	3	Sprint 1
US-02	As a user, I want to verify my mobile via OTP so my account is secure.	High	2	Sprint 1
US-03	As a user, I want to log in securely so only I can access my wallet.	High	3	Sprint 1
US-04	As a user, I want to see my wallet balance on the dashboard so I know my funds.	High	2	Sprint 1
US-05	As a user, I want to add funds to my wallet so I can make transfers.	Medium	2	Sprint 1
US-06	As a user, I want to send money to another user by entering their mobile number.	High	5	Sprint 2
US-07	As a user, I want to see a transaction receipt after every transfer.	High	3	Sprint 2
US-08	As a user, I want to see all past transactions in a list so I can track my spending.	Medium	3	Sprint 2
US-09	As a user, I want to filter transactions by date so I can find specific records.	Low	2	Sprint 2
US-10	As a user, I want to switch the app language to Hindi or Marathi so I can use it comfortably.	High	3	Sprint 3
US-11	As a user, I want to receive financial tips based on my spending so I can save better.	Medium	3	Sprint 3
US-12	As a user, I want to see a low-balance warning so I know when to add funds.	Medium	2	Sprint 3
US-13	As a user, I want the app to be mobile-friendly so I can use it on my phone.	High	2	Sprint 3
US-14	As an admin, I want to view all users and transactions so I can monitor the platform.	Low	3	Future

7. Definition of Done

A feature or sprint is considered 'Done' when all of the following criteria are met:

- Code is written, reviewed, and merged to the main branch
- Unit tests written and passing for the feature
- Feature is accessible and tested on a mobile browser
- No critical or high-severity bugs open
- Feature is demonstrated and accepted by all team members
- Relevant documentation updated

8. Agile Advantages Summary for MigrantPay

Agile Principle	How MigrantPay Applies It
Working software over documentation	We build a working prototype first; documentation is parallel, not blocking development
Responding to change	If evaluators suggest a new feature in Review 1, we can add it in Sprint 3 without restructuring the project
Continuous delivery	Each sprint produces a demonstrable, tested increment of the product
Team collaboration	Roles are clearly defined; standups every 3 hours ensure coordination
Simplicity	We focus on the MVP scope and avoid over-engineering
Sustainable pace	Work is divided so no single member is overloaded